



ΠΑΝΕΠΙΣΤΗΜΙΟ
ΙΩΑΝΝΙΝΩΝ

ΣΧΟΛΗ ΠΛΗΡΟΦΟΡΙΚΗΣ ΚΑΙ ΤΗΛΕΠΙΚΟΙΝΩΝΩΝ

ΤΜΗΜΑ

ΠΤΥΧΙΑΚΗ ΕΡΓΑΣΙΑ

**ΑΝΑΛΥΣΗ ΤΟΞΙΚΟΤΗΤΑΣ ΣΕ ΕΛΛΗΝΙΚΟ ΚΕΙΜΕΝΟ ΜΕ ΧΡΗΣΗ
ΜΗΧΑΝΙΚΗΣ ΜΑΘΗΣΗΣ**

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[Με πεζά στοιχεία και στοίχιση στο κέντρο. Περισσότερα του ενός ονόματα ακολουθούν σε διαφορετική γραμμή]

Επιβλέπων: Πέτρος Καρβέλης, Αναπληρωτής καθηγητής

Τόπος έκδοσης, Μήνας, Έτος ολοκλήρωσης

**TOXICITY ANALYSIS IN GREEK TEXT USING MACHINE
LEARNING ALGORITHMS**

Εγκρίθηκε από τριμελή εξεταστική επιτροπή

Τόπος, Ημερομηνία

ΕΠΙΤΡΟΠΗ ΑΞΙΟΛΟΓΗΣΗΣ

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Υπογραφή

ΕΥΧΑΡΙΣΤΙΕΣ

Είναι σημαντική η αναγνώριση της βοήθειας που έλαβε ο φοιτητής/ η φοιτήτρια κατά τη διάρκεια της προπαρασκευής της εργασίας του. Η βοήθεια μπορεί να είναι ακαδημαϊκή, τεχνική, γραμματειακή, διοικητική και προσωπική (π.χ. οικογένεια). Δεν υπερβαίνει τη μία παράγραφο.

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ΠΕΡΙΛΗΨΗ

Η τοξική συμπεριφορά στο διαδίκτυο όλο ένα και πολλαπλασιάζεται. Η μηχανική μάθηση μας προσφέρει εργαλεία για την αποτελεσματική ανίχνευση και πρόληψη τοξικών σχόλιων με την χρήση της Επεξεργασίας Φυσικής Γλώσσας. Τα δεδομένα που υπάρχουν για την εκμάθηση αλγορίθμων στον τομέα της τοξικότητας στα Ελληνικά είναι ελάχιστα σε σχέση με την πληθώρα δεδομένων στα αγγλικά. Σε αυτή την εργασία θα εξετάσουμε πως τρεις αλγόριθμοι μηχανικής μάθησης με διάφορα επίπεδα πολυπλοκότητας επεξεργάζονται δεδομένα τοξικότητας σε ελληνικό κείμενο και με πόση ακρίβεια κατηγοριοποιούν τα εν λόγω δεδομένα. Συγκεκριμένα θα χρησιμοποιήσουμε έναν βασικό αλγόριθμο με απλές μεθόδους, στην συνέχεια έναν ενδιάμεσο αλγόριθμο και τέλος τον πλέον σύγχρονο. Θα τους συγκρίνουμε και θα μελετήσουμε τα αποτελέσματα.

Λέξεις-κλειδιά: μηχανική μάθηση, επεξεργασία φυσικής γλώσσας, τοξικότητα, ελληνικά

ABSTRACT

Online toxicity is getting more common as time goes on. Machine learning provides us with effective tools for detecting and preventing online toxicity with the use of Natural Language Processing (NLP). Data used for training algorithms specifically in Greek is scarce in comparison to the plethora of data available for English. In this paper we will be examining how three machine learning algorithms with varying levels of complexity compute toxicity data in Greek text and how accurately they classify this data. Specifically, we will use a baseline algorithm using simple methods, then an intermediate algorithm with added complexity and finally the most ultramodern algorithm, we will compare them and interpret the results.

Keywords: machine learning, online toxicity, natural language processing, greek

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ΠΙΝΑΚΑΣ ΣΥΝΤΟΜΟΓΡΑΦΙΩΝ

NLP.....	Natural Language Processing
ΕΦΓ.....	Επεξεργασία Φυσικής Γλώσσας
BERT.....	Bidirectional Encoder Representations from Transformers
GRU.....	Gated Recurrent Unit
RNN.....	Recurrent Neural Network
OGTD.....	Offensive Greek Tweet Dataset
PTSD.....	Palo Toxicity and Sentiment Dataset
BiLSTM.....	Bidirectional Long Short-Term Memory
MLM.....	Masked Language Model
NSP.....	Next Sequence Prediction

GLOSSARY

[Διαγράψτε αυτή τη σελίδα αν δεν τη χρειάζεστε.]

INTRODUCTION

In the age of social media, the ways people interact on the internet and the content they consume has a direct impact on their day to day lives. Toxic behavior can transcend the boundaries of our screen and affect our psychological state. Online toxicity had also been correlated with increased probability of developing PTSD (Shmulewitz, 2025). We must promote healthy coexistence online and prevent mindless toxic behaviors. To achieve our goal, we can use the tools provided to us by the world of computer science and machine learning, specifically in the field of natural language processing.

Toxicity analysis in the Greek language is a complex problem since the data that is available is not only scarce, but many times doesn't include cursing or profanities commonly found in everyday casual speech by native Greek speakers, thus adding a challenge to the development and training of ML models. Additionally, the methods used by people online to spread hateful rhetoric are constantly evolving to outpace methods of detection. Methods include the use of embedded text in images and the use of ai for the generation of comments tailored to avoid detection and continue unscathed.

In this paper, we will dive into some of the most widely used ways to develop a machine learning model for natural language processing and observe the results they yield in the detection of toxicity in the Greek language.

1. Related Works

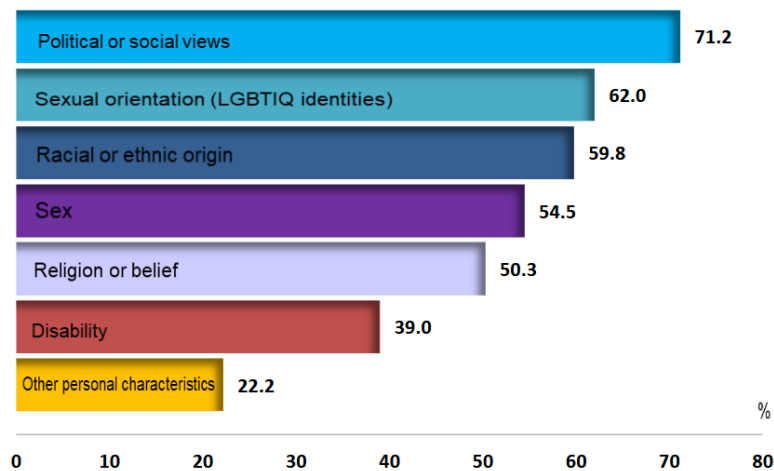
1.1 Toxicity analysis

Detecting toxic behavior online has been a task that has occupied researchers and professionals in the fields of IT and social media for many years. While many companies hire human content managers to detect and prevent toxic behavior, those exposed to this content Those exposed to this content may experience a decrease in their psychological health according to research by (Madriaza, 2025)

1.2 Toxicity analysis in Greek

The increasing use of social media in Greece makes it important to develop defense mechanisms against harmful online content and toxicity. According to (ELSTAT, 2023), the percentage of households with access to the internet increased from 56.3% in 2013 to 86.9% in 2023. Of the users who used the internet, 94.1% use it daily or almost daily. The same survey recorded for the first time the percentage of users who encountered degrading or hostile messages and news. At least 30% of users encountered some kind of toxicity for various reasons listed in the graph below.

Graph 9. Degrading online messages by reason. Percentage distribution of population having encountered degrading messages / news in blogs / vlogs / social media / news sites. 1st quarter 2023



Significant research efforts have been made to address this problem. In the work “Offensive Language Identification in Greek” (Zeses Pitenis, 2020), a special dataset was developed for detecting toxicity in the Greek language with 4,779 Greek toxic and non-toxic tweets (OGTD). The thesis “Toxicity Detection on Greek Tweets” (Theodoros, 2021)

used OGTD for this specific project as well as the Palo Toxicity and Sentiment Dataset (PTSD) provided by PALOServices, which consists of 1,452 samples and includes deeper categorization such as toxicity type, sentiment, and assessment. In the study “Multimodal Hate Speech Detection in Greek Social Media” (Perifanos & Goutsos, 2021), OGTD was used along with two other publicly available datasets, while a new dataset consisting of 4,000 tweets was developed. In these three studies, it is observed that the most popular technology is deep learning models, with the work “Offensive Language Identification in Greek” also using traditional machine learning as a basic method for comparison.

1.3 Methods and models

To avoid exposure to harmful content, multiple methods have been developed using modern techniques and technologies. BERT models are extremely popular in this field. For example, the study “Multimodal Hate Speech Detection in Greek Social Media” (Perifanos & Goutsos, 2021) used publicly available models such as the Greek BERT model as well as a modification of a RoBERTa language model for the Greek language. Furthermore, the work “Toxicity Detection on Greek Tweets” (Theodoros, 2021) presented the use of three BERT models and a GRU model with Greek FastText embeddings. The aforementioned models are classified as deep learning models and show excellent efficiency in training for complex tasks. In the work “Offensive Language Identification in Greek”, as a measure of comparison, TF-IDF and logistic regression were applied, a practice considered common in the field. Therefore, it is observed that, despite the small use of classical machine learning, deep learning models dominate the field.

Models like BERT and RoBERTa are based on a transformer architecture, meaning they use transformers. Transformers are a type of neural network architecture that transforms or modifies an input sequence into an output sequence. This is achieved through learning the environment and monitoring the relationships between the elements of the sequence.

1.4 Challenges and limitations

The Greek language presents many peculiarities that create obstacles for the task of toxicity detection. Intonation is a characteristic that few languages express in writing and Greek is one of them. When communicating on the internet and especially on social media, an informal style prevails, so it is common to omit the accent in text. However, this can cause inaccuracies in machine learning, as a letter with an accent and the same letter without an accent are treated as different characters. Additionally, Greeks have a wide

variety of swear words and insults which may be based on their region of origin within Greece, which makes them rare in use, especially in written language.

The problem of lack of data occurs for many languages whose speaker population is relatively small. Greek is considered a low-resource language in the field of Natural Language Processing (NLP) and thus finding data is already difficult. However, the peculiarities and variety of the language further complicate the achievement of the goal. Greek swear words are numerous and are not easy to record in a standardized manner. Therefore, combined with the addition or absence of accents and orthographic variations, the processing of Greek becomes extremely difficult.

2. Datasets

2.1 Offensive Greek Tweet Dataset

OGTD v1.0 was created between May and June 2019. The Twitter API was used to collect tweets. Tweets containing popular hashtags from Greek television programs were collected through the Twitter API. Municipal and European elections were being held at the same time, which would likely have generated an increased number of complaints and obscene comments, making the period suitable for data collection. Additionally, the search with the keyword "you are" was utilized with the assumption that some swearing would follow.

The total number of tweets was 49,154 and the next step was pre-processing. Emoji, URLs and emoticons were removed from all tweets and user references were replaced with @USER. The use of double punctuation such as question marks and exclamation marks was also normalized. After preprocessing and deleting duplicate tweets, the number was reduced to 46,218 from which 5,000 were randomly selected for annotation.

Light-Tag¹ was used for annotation. Annotations where there was 100% agreement from the software were considered acceptable and added to the total. For annotations where there was not 100% agreement, The annotation that prevailed with a percentage greater than 66% was selected. When there was absolute disagreement, intervention by human assistants was required. In the end, the total consisted of 4,779 tweets with 29% of them considered offensive, as shown in the table.

Labels	Training Set	Test Set	Total
Offensive	955	446	1.401
Not Offensive	2.390	988	3.378
Total	3.345	1.434	4.779

Table 1: Distribution of labels in the OGTD v1.0

¹ <https://www.linkedin.com/company/lighttag/?originalSubdomain=de>

2.2 Greek FastText word-vectors

Meta's (formerly known as Facebook) fastText was first introduced in the research paper "Enriching Word Vectors with Subword Information" by (Bojanowski, Grave, Joulin, & Mikolov, 2017). fastText is a free, lightweight, open-source library that allows users to learn text representations and text classifiers.

FastText extends the Skip-gram and CBOW models by representing words as units of n-grams of characters instead of individual units. This fundamental shift allows the model to create embeddings for previously unseen words and capture morphological relationships between related terms.

The Facebook AI Research (FAIR) lab offers pre-trained word embeddings in English as well as 157 other languages, including Greek. These models were trained using CBOW with positional weights, in dimension 300, with n-grams of length 5 characters, a window size of 5, and 10 negatives.

3. Methods Used

3.1 Pre-Processing

The OGTD data includes tweets in Greek with uppercase and lowercase letters as well as word stress. The creators of the dataset pre-processed the tweets by removing useless information as mentioned in [2.1](#) by deleting links, emojis and emoticons.

A human reads the word with and without an accent and considers them the same, but in the case of computers a stressed character is treated as different from the same character without an accent. To achieve higher accuracy while avoiding confusion, it was decided to process the dataset by first removing accents from all words that contained accents.

In addition to accents, another post-processing method is to remove capitalization from letters. Therefore, as with accents, the data was processed by converting all uppercase letters to lowercase.

After this processing, to determine if it was necessary or if there was no difference in the result, all the algorithms developed were run with the data before and after processing. The results are presented in the table below.

	Without pre-processing	With pre-processing
Baseline model	86	89
Intermediate model	57	85
Advance model	82	84

Table 2: Macro-Averaged F1 score before and after pre-processing

Looking at the results, we notice that after preprocessing there is an increase in the macro-avg score of 2-3% for each model. The most drastic change however, can be observed with our intermediate model with a rise from 57% to 85% proving how big a change this pre-processing methods actually makes.

3.2 Baseline model with TF-IDF vectorizer and binomial logistic regression

The first and baseline model in this paper uses a TF-IDF vectorizer for the vectorization of words. Term frequency (TF) measures the frequency with which a word appears in a document. Since a term can appear more times in a longer document and fewer times in a shorter document, the term frequency is often normalized by the length of the document.

$$TF(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}$$

Inverse document frequency (IDF) measures how important a term is. However, terms like "is", "and", and "are" might be more frequent but not particularly significant. This technique assigns more weight to rare terms while assigning less weight to frequent ones.

$$IDF(t, D) = \log \left(\frac{N}{|\{d \in D: t \in d\}|} \right)$$

where:

- N is the total number of documents in the corpus D .
- $|\{d \in D: t \in d\}|$ is the number of documents where the term t appears.
- If the term is not in the corpus, this will lead to a division-by-zero. It is therefore common to adjust the denominator to $1 + |\{d \in D: t \in d\}|$.

When transforming text to vectors using TF-IDF, each document is represented as a vector. Each dimension of the vector corresponds to a separate term from the corpus vocabulary. The value in each dimension is the TF-IDF score of the term in the document. To handle the vast size of the vocabulary and the sparsity of individual document vectors, implementation usually use sparse matrix representations.

After the model completes word vectorization, it uses the resulting word vectors as input and computes a weight for each feature:

$$z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

Where:

- x_i : TF-IDF value for word i
- w_i : learned importance of that word

The computed weight is fed as input into a sigmoid function.

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The output of the sigmoid function ranges between 0 and 1. In this case, **offensive** is labeled as 1 and **not offensive** is labeled as 0. Based on the output of the function, the model determines whether the selected tweet is closer to being offensive or not offensive.

This model uses no deep learning as a means of measuring the capabilities of classic machine learning algorithms in our task.

3.3 Intermediate model with FastText and BiLSTM

As explained previously in section [2.2](#), this Model uses Meta's FastText library to import Greek word embeddings. After importing the word embeddings, the intermediate model employs a Bidirectional Long Short-Term Memory (BiLSTM) classifier on our training and testing data.

BiLSTM is a type of Recurrent Neural Network (RNN). Unlike traditional RNNs that process input sequences in only one direction (either forward or backward), BiLSTM processes the sequence in both directions simultaneously. It consists of two LSTM layers: one processing the sequence in the forward direction and the other in the backward direction. Each layer maintains its own hidden states and memory cells.

During the forward pass, the input sequence is fed into the forward LSTM layer from the first time step to the last. At each time step, the forward LSTM computes its hidden state and updates its memory cell based on the current input and the previous hidden state and memory cell.

Simultaneously, the input sequence is also fed into the backward LSTM layer in reverse order, from the last time step to the first. Similar to the forward pass, the backward LSTM

computes its hidden state and updates its memory cell based on the current input and the previous hidden state and memory cell.

Once the forward and backward passes are complete, the hidden states from both LSTM layers are combined at each time step. This combination can be as simple as concatenating the hidden states or applying some other transformation.

The advantage of BiLSTM is that it captures not only the context preceding a specific time step (as in traditional RNNs) but also the context that follows. By considering both past and future information, BiLSTM can capture richer dependencies in the input sequence.

Additionally, the Adam optimizer is also utilized in this model. Adam optimization was introduced in the International Conference on Learning Representations and later published as a conference paper in Adam: A method for stochastic optimization (Kingma & Ba, 2015). Adam is a combination of two other optimization algorithms, RMSProp and AdaGrad.

Finally, our intermediate model uses binary cross-entropy loss. Binary Cross-Entropy Loss is a widely used loss function in binary classification problems. For a dataset with N instances, the Binary Cross-Entropy Loss is calculated as:

$$BCE = -\frac{1}{N} \sum_{i=1}^N (y_i \cdot \log(p_i) + (1 - y_i) \log(1 - p_i))$$

Where:

- N is number of samples
- y_i true label for sample i (0 or 1)
- p_i model-predicted probability for class 1 for sample i

3.3 Advanced model with Greek BERT

BERT, short for Bidirectional Encoder Representations from Transformers, is a Machine Learning model for natural language processing. It was introduced in the paper “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding” (Devlin, Chang, Lee, & Toutanova, 2018) and serves as a swiss army knife solution to 11+ of the most common language tasks, such as sentiment analysis and named entity recognition.

BERT, as the name explains, is a multi-layer bidirectional Transformer encoder and was trained on a massive dataset of 3.3 billion words. Specifically, BERT was trained on Wikipedia (~2.5B words) and Google's BookCorpus (~800M words). BERT is pre-trained with two tasks. First task "Masked LM" (MLM) in which some percentage of the words in the input are hidden(masked) and the model tries to predict them based on their context. This is also referred to as the Cloze task in literature. Second task is "Next Sentence Prediction" (NSP) which helps the model understand the relationship between sentences, for example if one sentence follows another sentence or not.

BERT is a transformer-based model. The Transformer architecture makes it possible to parallelize ML training extremely efficiently. Massive parallelization thus makes it feasible to train BERT on large amounts of data in a relatively short period of time. Transformers use an attention mechanism to observe relationships between words. A concept originally proposed in the popular paper "Attention Is All You Need" (Vaswani, et al.) sparked the use of Transformers in NLP models all around the world.

Transformers work by leveraging attention, a powerful deep-learning algorithm, first seen in computer vision models. Machine Learning models need to learn how to pay attention only to the things that matter and not waste computational resources processing irrelevant information. Transformers create differential weights signaling which words in a sentence are the most critical to further process. A transformer does this by successively processing an input through a stack of transformer layers, usually called the encoder. If necessary, another stack of transformer layers (the decoder) can be used to predict a target output. BERT, however, doesn't use a decoder. Transformers are uniquely suited for unsupervised learning because they can efficiently process millions of data points.

For our task, which is in Greek, the original BERT model wouldn't be able to recognize the language so we used another model based on the original. GreekBERT was created using the official code provided in google's BERT github repository with a similar size and parameters to the base BERT model (12-layer, 768-hidden, 12-heads, 110M parameters). GreekBERT was trained on the Greek part of Wikipedia, The Greek part of European Parliament Proceedings Parallel Corpus and The Greek part of OSCAR, a cleansed version of Common Crawl. It is clear to see that this much better fits our purpose.

4.0 Results

ΠΑΡΑΡΤΗΜΑ

[Υλικό που είναι ογκώδες, παραδείγματος χάριν αποτελέσματα ερωτηματολογίων, σχήματα, πίνακες κ.ά. που εμποδίζουν τον αναγνώστη να κατανοήσει τη συνέχεια του κειμένου, μπορεί να τοποθετηθεί σε παραρτήματα. Η αρίθμηση των παραρτημάτων γίνεται με κεφαλαία ελληνικά γράμματα Α, Β, Γ, ... ενώ σχήματα, σχέδια, πίνακες κ.λπ., που περιλαμβάνονται πρέπει να ονομάζονται Σχήμα Α1, Πίνακας Β2, κ.λπ.]

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[Οπισθόφυλλο. Κενή σελίδα]